Gayathri Gunda

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Summary

AI/ML-focused B.Tech student building agentic RL environments and LLM systems. Trained a 1.5B model with REINFORCE + LoRA to 34% held-out task success where a frozen 3B baseline scored 0%, and ship production-style services with FastAPI, Docker, and full-stack tooling.

Education

Gokaraju Rangaraju Institute of Engineering and Technology (GRIET) Hyderabad, India
B.Tech in Computer Science & Engineering (Artificial Intelligence) *2023 – 2027 (Expected)*

- CGPA: 7.98/10

Projects

MCPArena – Adversarial MCP RL Environment
Python, PyTorch, REINFORCE, LoRA, Unsloth, FastAPI, Docker

- Engineered an OpenEnv environment that places an LLM agent in a hostile MCP tool catalog (misleading docstrings, prompt injection in tool outputs, decoy leak-sink tools, hard cost budget), rotating the catalog every episode to force genuine tool reading over name memorization.
- Trained Qwen2.5-1.5B with REINFORCE + LoRA and reward shaping to reach **34%** held-out task success on unseen tools at 3× adversarial training density, versus **0%** for a frozen 3B baseline; cut malformed tool-call errors by **83%** and broke a no-commit failure mode through reward-shaping design.
- Designed a five-rubric reward (task success, cost, adversarial avoidance, calibration, well-formedness); deployed on HF Spaces via FastAPI + Docker with a validator-passing OpenEnv API and published run logs as a public HF dataset.

PromptGuard – Prompt-Injection Defense Environment
Python, FastAPI, Pydantic, Docker

- Built an OpenEnv-compatible RL environment where agents craft system prompts to defend against prompt-injection attacks spanning six categories (including override, role-play, obfuscation, indirect injection, and social engineering) across three difficulty tiers.
- Implemented composite scoring that balances defense rate against task utility, so agents cannot win by refusing everything; served through an OpenAI-compatible LLM interface with FastAPI, Pydantic, and Docker.

Technical Skills

Languages	Python, TypeScript, JavaScript, Java, Go
ML & RL	PyTorch, Hugging Face (Transformers, Datasets, Spaces), LoRA / Unsloth, REINFORCE and policy-gradient methods, reward shaping, OpenEnv
LLM Systems	Model Context Protocol (MCP), agent loops, RAG (pgvector, cosine similarity), structured extraction (Zod), prompt-injection defense, OpenAI, Fireworks AI, Azure OpenAI, Google Cloud STT/TTS
Backend & Frameworks	FastAPI, Pydantic, Express, tRPC, Spring Boot, Streamlit
Data & Infrastructure	PostgreSQL (pgvector), MongoDB, Redis, SQLite, Docker, Kubernetes, GitHub Actions, AWS (EC2, S3, Elastic Beanstalk), Cloudflare R2

Leadership Experience

Event Manager Lead 2024 – 2026
Technical Club, GRIET

- Led end-to-end execution of **2 college hackathons** with **100+ participants**: timelines, registrations, resource allocation, and day-of scheduling.
- Coordinated project evaluations and presentations between **15+ student teams** and faculty judges.

Certifications

- AWS Educate Machine Learning Certificate – Amazon Web Services
- Introduction to Data Analytics – IBM / Coursera
- IEEE AI & Big Data Certificate – IEEE